251103 Phasor domain

Sinusoidal Signals:

Form: $v(t) = V_m \cos\left(\frac{2\pi t}{T} + \phi\right)$

→ Period vs. Frequency:

$$\boxed{T = \frac{1}{f} \iff f = \frac{1}{T}}$$

→ Angular frequency Vs. Frequency

$$\boxed{\omega = 2\pi f = \frac{2\pi}{T}}$$

→ Phase Shift Vs. Time Shift

$$\boxed{\Delta t = \left(\frac{\phi}{2\pi}\right) T = \frac{\phi}{2\pi f} = \phi/\omega}$$

*Phases are always relative to a reference

ex/ $v(t) = V_m \cos(\omega t + \phi)$ leads the reference $v_{ref}(t) = \cos(\omega t)$ by ϕ.

(If ϕ were negative then it would be a lag.)

Complex Algebra:

→ Rectangular form

$$\boxed{Z = X + jY}$$

where $j = \sqrt{-1}$,
$Re(z) = X$, and $Im(z) = Y$

→ Polar form

$$\boxed{Z = |z| e^{j\phi} = |z| \angle \phi}$$

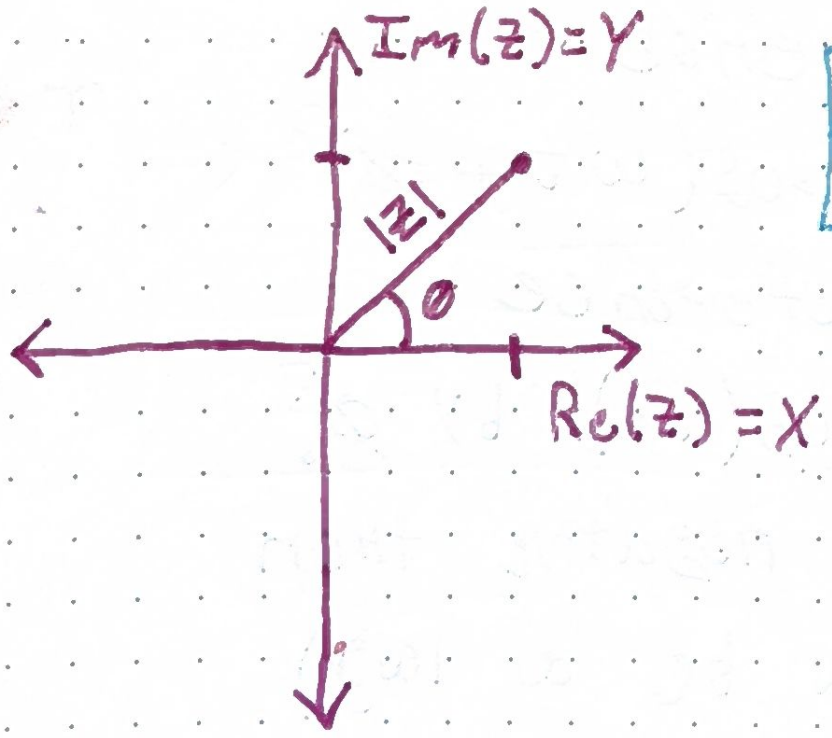
→ Euler's identity

$$\boxed{e^{j\phi} = \cos(\phi) + j \sin(\phi)}$$

$\Rightarrow Z = |z| \cos(\phi) + j|z| \sin(\phi)$

$\Rightarrow \boxed{Re(z) = |z| \cos(\phi) = X}$

$\Rightarrow \boxed{Im(z) = |z| \sin(\phi) = Y}$



$$\boxed{|z| = \pm\sqrt{X^2 + Y^2}}$$

$$\boxed{\phi = \tan^{-1}(Y/X)}$$

$$\boxed{e^{j\pi} = -1}$$

Phasor Domain

→ Time-domain

* $\boxed{x(t) = Re(}$

* $\boxed{\tilde{X} = |\tilde{X}| e^{j\phi}}$

Take $v(t) = V_0$

$\Rightarrow v(t) = Re[$

Since $V_0 e^{j(\omega}$

$\Rightarrow v(t) = Re[$

$\boxed{\therefore v(t) = V_0 \cos(}$

If $\phi = 0$

$\Rightarrow v(t) = V_0 \cos$

$j = \sqrt{-1}$,

$= X$, and $Im(Z) = Y$

(θ)

$= \pm\sqrt{X^2 + Y^2}$

$= \tan^{-1}(Y/X)$

$^{\pi} = -1$

Phasor Domain:

→ Time-domain / Phasor-domain

$$\boxed{x(t) = Re(\tilde{X} e^{j\omega t})}$$

Phasor

Angular Frequency

$$\boxed{\tilde{X} = |\tilde{X}| e^{j\phi}}$$

Phase

Amplitude

Take $v(t) = V_0 \cos(\omega t + \phi)$

$\Rightarrow v(t) = Re\left[V_0 e^{j(\omega t + \phi)}\right]$

Since $V_0 e^{j(\omega t + \phi)} = V_0 \cos(\omega t + \phi) + j V_0 \sin(\omega t + \phi)$

$\Rightarrow v(t) = Re\left[\underbrace{V_0 e^{j\phi}}_{\text{The Phasor}} e^{j\omega t}\right]$

$$\boxed{\therefore v(t) = V_0 \cos(\omega t + \phi) \Longleftrightarrow \tilde{V} = V_0 e^{j\phi}}$$

If $\phi = 0$

$\Rightarrow v(t) = V_0 \cos(\omega t) \Longleftrightarrow \tilde{V} = V_0$

Consider the time derivative of $i(t)$

where $i(t) = Re[\tilde{I} e^{j\omega t}]$

$$\Rightarrow \frac{di}{dt} = \frac{d}{dt} Re[\tilde{I} e^{j\omega t}] = Re[\underbrace{j\omega \tilde{I}}_{\text{phasor}} e^{j\omega t}]$$

$$\boxed{\therefore \frac{di(t)}{dt} \Leftrightarrow j\omega \tilde{I}}$$

Similarly: $\boxed{\int i(t)\, dt \Leftrightarrow \tilde{I}/j\omega}$

Impedance:

*Ohm's Law $\boxed{v_R = i_R R \Leftrightarrow \tilde{V}_z = \tilde{I}_z Z}$

→ Inductors

$$v_L = L\frac{di_L}{dt} \Rightarrow \tilde{V}_L = L\frac{d}{dt}\tilde{I}_L$$

$$\Rightarrow \tilde{V}_L = \underbrace{j\omega L}_{\text{Impedance}} \tilde{I}_L$$

$$\boxed{\therefore Z_L = j\omega L}$$

→ Capacitors

$$i_c = C\frac{dv_c}{dt} \Rightarrow$$

$$\Rightarrow \tilde{V}_c = \underbrace{\frac{1}{j\omega C}}$$

$$\boxed{\therefore Z_c = \frac{1}{j\omega C}}$$

→ Resistors

$$\boxed{Z_R = R}$$

*Notice Z_L

↳ But the Ph

↳ $\angle Z_L = \pi/2$

Where: $\tilde{V} = \tilde{I} Z$

$\Rightarrow \boxed{\angle Z = \angle}$

∴ For:

→ Inductor

Current

→ Capacitor:

Current

→ Resistor: Volt

ive of $\dot{i}(t)$

]

$[j\omega \tilde{I} e^{j\omega t}]$

phasor

$\boxed{\tilde{I}/j\omega}$

$\tilde{I}_L$

$\boxed{Z}$

e

→ Capacitors

$$\dot{i}_c = C\frac{dV_c}{dt} \Rightarrow \tilde{I}_c = C\frac{d}{dt}\tilde{V}_c \Rightarrow \tilde{I}_c = j\omega C\tilde{V}_c$$

$$\Rightarrow \tilde{V}_c = \underbrace{\frac{1}{j\omega C}}_{\text{Impedance}}\tilde{I}_c$$

$$\boxed{\therefore Z_c = \frac{1}{j\omega C}}$$

→ Resistors

$$\boxed{Z_R = R}$$

* Notice Z_L and Z_c depend on ω
  - ↳ But the phases do not depend on ω
  - ↳ $\angle Z_L = \pi/2$ rad and $\angle Z_c = -\pi/2$

Where: $\tilde{V} = \tilde{I}Z \Rightarrow \boxed{Z = \tilde{V}/\tilde{I}}$

$\Rightarrow \boxed{\angle Z = \angle\tilde{V} - \angle\tilde{I}}$

∴ For:

→ Inductor: Voltage leads current by $\pi/2$

→ Capacitor: Voltage lags current by $\pi/2$

→ Resistor: Voltage and current same phase